

Title of the Presentation

The subtitle of the presentation

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Example

Overview

- 1 First Section
- 2 Second Section
- 3 Third Section
- 4 References

Full Frame Title

subtitle

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$$E = mc^2 \tag{1}$$

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¹ Alpha

² Beta
Gamma

Second Frame

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- ▶ ABC
- ▶ ABC
 - DEF
 - * XYZ
 - * XYZ
 - DEF
- ▶ ABC

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Second Frame

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- ▶ ABC
- ▶ ABC
 - DEF
 - * XYZ
 - * XYZ
 - DEF
- ▶ ABC

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Third Frame

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1. ABC

2. ABC

(a) DEF

(b) DEF

i. XYZ

ii. XYZ

3. ABC

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Theorem

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Comparison of Moduli Spaces

Space	Dimension	Compact?
M_g (smooth)	$3g - 3$	No
$\overline{M}_g$ (stable)	$3g - 3$	Yes
$M_{g,n}$ (marked)	$3g - 3 + n$	No
$\overline{M}_{g,n}$	$3g - 3 + n$	Yes

Table: Dimensions of moduli spaces

Example: $g = 1$ (Elliptic Curves)

For genus 1 curves with a marked point:

- ▶ $M_{1,1} \cong \mathcal{H} / \mathrm{SL}(2, \mathbb{Z})$
- ▶ Dimension = 1
- ▶ Compactification adds nodal cubic

Remark

The boundary $\overline{M}_g \setminus M_g$ consists of nodal curves with lower genus components.

References

- [1] **S. W. Hawking**, *Particle creation by black holes Particle creation by black holes Particle creation by black holes*, Commun. Math. Phys. 43, 199–220 (1975).
- [2] **E. Witten**, *Search for a realistic Kaluza-Klein theory*, Nucl. Phys. B 186, 412-428 (1981).

Thank You!